In [1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```python
data = pd.read_csv("QVI_data.csv")
```

In [3]:
```python
data["DATE"] = pd.to_datetime(data["DATE"])
```

In [4]:
```python
data["YEARMONTH"] = data["DATE"].dt.year * 100 + data["DATE"].dt.month
```

In [5]:
```python
print(data["YEARMONTH"].unique())
```

```
[201810 201809 201903 201811 201812 201807 201906 201904 201901 201808
 201905 201902]
```

In [6]:
```python
print(data["STORE_NBR"].nunique(), "stores")
```

```
272 stores
```

Monthly metrics per store banao

In [7]:
```python
monthly_metrics = data.groupby(["STORE_NBR", "YEARMONTH"]).agg(
    TOT_SALES=("TOT_SALES", "sum"),
    N_CUSTOMERS=("LYLTY_CARD_NBR", "nunique"),
    N_TXNS=("TXN_ID", "nunique"),
    N_TXN_PER_CUST=("TXN_ID", "nunique")  # temp, will fix below
).reset_index()
```

In [8]:
```python
# Average transactions per customer
monthly_metrics["TXN_PER_CUST"] = monthly_metrics["N_TXNS"] / monthly_metrics["N
```

In [9]:
```python
print(monthly_metrics.shape)
```

```
(3169, 7)
```

In [10]:
```python
print(monthly_metrics.head(10))
```

```
   STORE_NBR  YEARMONTH  TOT_SALES  N_CUSTOMERS  N_TXNS  N_TXN_PER_CUST  \
0          1     201807      206.9           49      52              52
1          1     201808      176.1           42      43              43
2          1     201809      278.8           59      62              62
3          1     201810      188.1           44      45              45
4          1     201811      192.6           46      47              47
5          1     201812      189.6           42      47              47
6          1     201901      154.8           35      36              36
7          1     201902      225.4           52      55              55
8          1     201903      192.9           45      49              49
9          1     201904      192.9           42      43              43

   TXN_PER_CUST
0      1.061224
1      1.023810
2      1.050847
3      1.022727
4      1.021739
5      1.119048
6      1.028571
7      1.057692
8      1.088889
9      1.023810
```

In [11]:
```python
months_per_store = monthly_metrics.groupby("STORE_NBR")["YEARMONTH"].count()
full_stores = months_per_store[months_per_store == 12].index
```

In [12]:
```python
print("Stores with full 12 months:", len(full_stores))
```

```
Stores with full 12 months: 260
```

In [13]:
```python
monthly_metrics_full = monthly_metrics[monthly_metrics["STORE_NBR"].isin(full_st
print(monthly_metrics_full.shape)
```

```
(3120, 7)
```

In [14]:
```python
print([s for s in [77, 86, 88] if s in full_stores])
```

```
[77, 86, 88]
```

In [15]:
```python
pretrial_months = [201807,201808,201809,201810,201811,201812,201901]
```

In [16]:
```python
pretrial = monthly_metrics_full[monthly_metrics_full["YEARMONTH"].isin(pretrial_
```

In [17]:
```python
def calculate_correlation(pretrial_df, metric_col, trial_store):
    trial_data = pretrial_df[pretrial_df["STORE_NBR"] == trial_store].set_index(

    results = []
    for store in pretrial_df["STORE_NBR"].unique():
        if store == trial_store:
            continue
        control_data = pretrial_df[pretri
        al_df["STORE_NBR"] == store].set_index("YEARMONTH")[metric_col]
        corr = trial_data.corr(control_data)
        results.append({"STORE_NBR": store, "corr_measure": corr})

    return pd.DataFrame(results)
```

In [18]:
```python
corr_sales_77 = calculate_correlation(pretrial, "TOT_SALES", 77)
print(corr_sales_77.sort_values("corr_measure", ascending=False).head(10))
```

```
     STORE_NBR  corr_measure
67          71      0.914106
220        233      0.903774
110        119      0.867664
15          17      0.842668
2            3      0.806644
38          41      0.783232
46          50      0.763866
148        157      0.735893
153        162      0.729740
243        257      0.724927
```

Magnitude distance function

In [25]:
```
def calculate_magnitude_distance(pretrial_df, metric_col, trial_store):
    trial_data = pretrial_df[pretrial_df["STORE_NBR"] == trial_store].set_index(
    results = []
    for store in pretrial_df["STORE_NBR"].unique():
        if store == trial_store:
            continue
        control_data = pretrial_df[pretrial_df["STORE_NBR"] == store].set_index(
        abs_diff = (trial_data - control_data).abs().mean()
        results.append({"STORE_NBR": store, "abs_diff": abs_diff})
    dist_df = pd.DataFrame(results)
    min_d = dist_df["abs_diff"].min()
    max_d = dist_df["abs_diff"].max()
    dist_df["mag_measure"] = 1 - (dist_df["abs_diff"] - min_d) / (max_d - min_d)
    return dist_df
```

In [26]:
```
mag_sales_77 = calculate_magnitude_distance(pretrial, "TOT_SALES", 77)
print(mag_sales_77.sort_values("mag_measure", ascending=False).head(10))
```

```
     STORE_NBR   abs_diff  mag_measure
220        233  18.828571     1.000000
241        255  28.571429     0.991889
179        188  31.100000     0.989784
49          53  32.471429     0.988642
122        131  33.042857     0.988167
42          46  33.228571     0.988012
202        214  34.442857     0.987001
46          50  34.714286     0.986775
195        205  36.057143     0.985657
178        187  37.685714     0.984301
```

Correlation + Magnitude

In [27]:
```
def get_control_store(pretrial_df, metric_col, trial_store, weight_corr=0.5, wei
    corr_df = calculate_correlation(pretrial_df, metric_col, trial_store)
    mag_df = calculate_magnitude_distance(pretrial_df, metric_col, trial_store)

    combined = corr_df.merge(mag_df, on="STORE_NBR")
    combined["score"] = weight_corr * combined["corr_measure"] + weight_mag * cc

    return combined.sort_values("score", ascending=False)

scores_77 = get_control_store(pretrial, "TOT_SALES", 77)
print(scores_77.head(10))
```

```
     STORE_NBR  corr_measure    abs_diff  mag_measure     score
220        233      0.903774   18.828571     1.000000  0.951887
38          41      0.783232   45.342857     0.977927  0.880579
46          50      0.763866   34.714286     0.986775  0.875320
15          17      0.842668  155.614286     0.886125  0.864397
107        115      0.689159   87.657143     0.942700  0.815929
158        167      0.657110   53.485714     0.971148  0.814129
251        265      0.639759   48.271429     0.975489  0.807624
221        234      0.696325  143.685714     0.896056  0.796190
49          53      0.532764   32.471429     0.988642  0.760703
78          84      0.684348  219.928571     0.832583  0.758465
```

Multi-metric combined score

In [28]:
```
def get_control_store_multi(pretrial_df, trial_store, metrics=["TOT_SALES","N_CU
    all_scores = None
    for metric in metrics:
        scores = get_control_store(pretrial_df, metric, trial_store)[["STORE_NBR
        scores = scores.rename(columns={"score": f"score_{metric}"})
        if all_scores is None:
            all_scores = scores
        else:
            all_scores = all_scores.merge(scores, on="STORE_NBR")

    score_cols = [c for c in all_scores.columns if c.startswith("score_")]
    all_scores["final_score"] = all_scores[score_cols].mean(axis=1)
    return all_scores.sort_values("final_score", ascending=False)

control_77 = get_control_store_multi(pretrial, 77)
print(control_77.head(10))
```

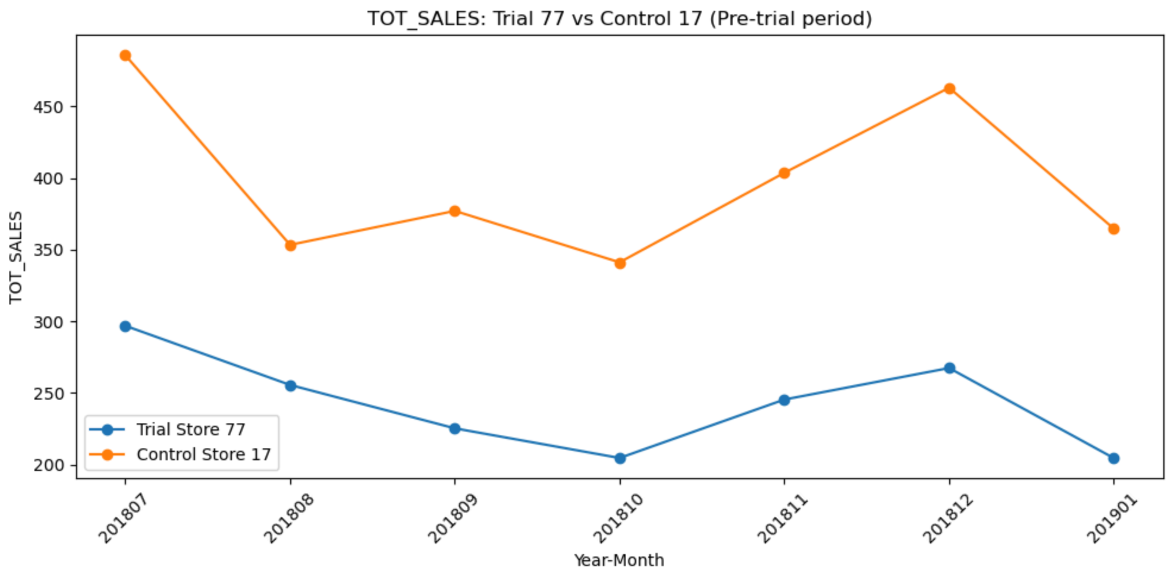
```
C:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\_function_base_impl.py:2999:
RuntimeWarning: invalid value encountered in divide
  c /= stddev[:, None]
C:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\_function_base_impl.py:3000:
RuntimeWarning: invalid value encountered in divide
  c /= stddev[None, :]
```

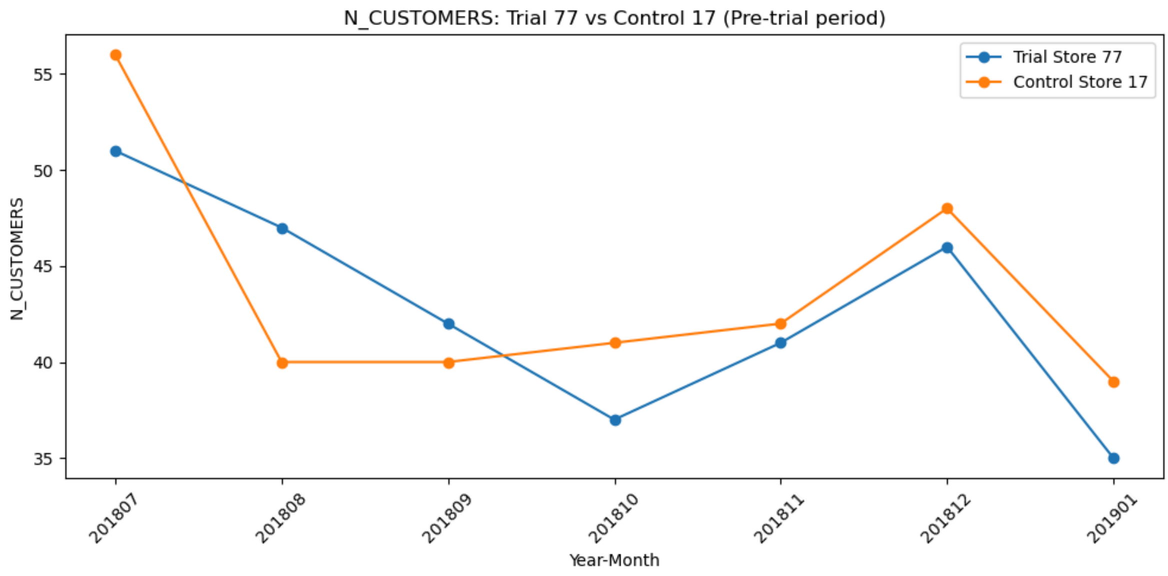
```
    STORE_NBR  score_TOT_SALES  score_N_CUSTOMERS  score_TXN_PER_CUST  \
3          17         0.864397           0.858222            0.684392
9          84         0.758465           0.893020            0.670811
20        145         0.689454           0.820516            0.782399
4         115         0.815929           0.844780            0.623858
29         37         0.655673           0.711514            0.856030
33        248         0.631211           0.807019            0.730719
0         233         0.951887           0.995179            0.221718
5         167         0.814129           0.836580            0.493785
24        176         0.673940           0.509122            0.927979
7         234         0.796190           0.766486            0.539961

    final_score
3      0.802337
9      0.774099
20     0.764123
4      0.761523
29     0.741072
33     0.722983
0      0.722928
5      0.714831
24     0.703680
7      0.700879
```

In [29]:
```
print(pretrial[pretrial["STORE_NBR"]==233]["TXN_PER_CUST"])
print(pretrial.groupby("STORE_NBR")["TXN_PER_CUST"].std().sort_values().head(10)
```

```
2699    1.058824
2700    1.041667
2701    1.071429
2702    1.028571
2703    1.025000
2704    1.063830
2705    1.000000
Name: TXN_PER_CUST, dtype: float64
STORE_NBR
244    0.0
61     0.0
52     0.0
42     0.0
127    0.0
139    0.0
224    0.0
99     0.0
258    0.0
198    0.0
Name: TXN_PER_CUST, dtype: float64
```

In [30]:
```
def plot_trial_vs_control(pretrial_df, metric_col, trial_store, control_store):
    trial_data = pretrial_df[pretrial_df["STORE_NBR"]==trial_store].sort_values(
    control_data = pretrial_df[pretrial_df["STORE_NBR"]==control_store].sort_val

    plt.figure(figsize=(10,5))
    plt.plot(trial_data["YEARMONTH"].astype(str), trial_data[metric_col], marker
    plt.plot(control_data["YEARMONTH"].astype(str), control_data[metric_col], ma
    plt.title(f"{metric_col}: Trial {trial_store} vs Control {control_store} (Pr
    plt.xlabel("Year-Month")
    plt.ylabel(metric_col)
    plt.legend()
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()

plot_trial_vs_control(pretrial, "TOT_SALES", 77, 17)
plot_trial_vs_control(pretrial, "N_CUSTOMERS", 77, 17)
```



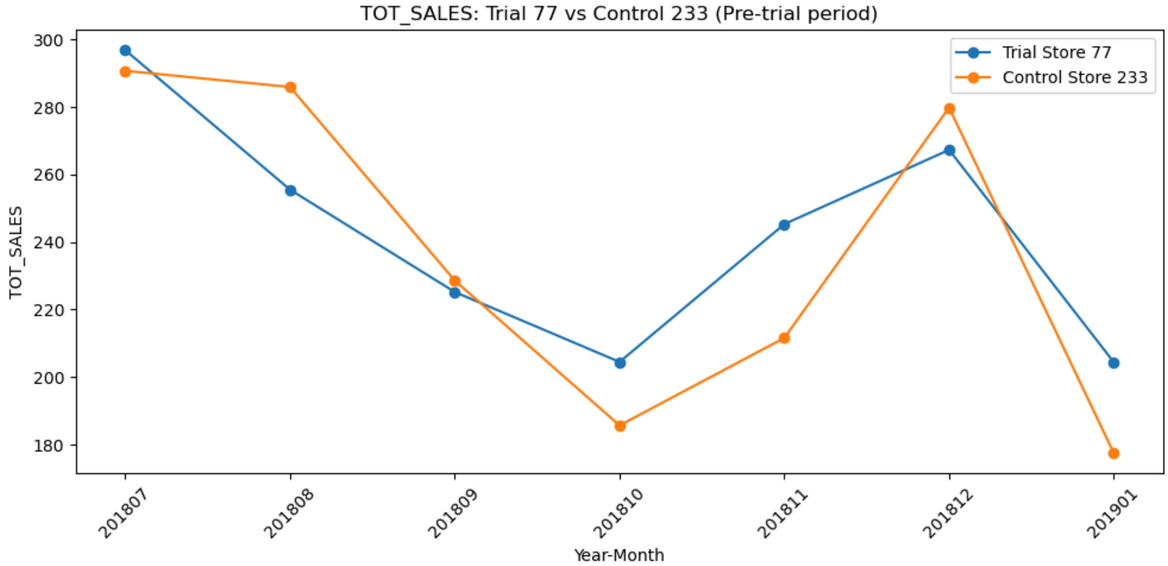


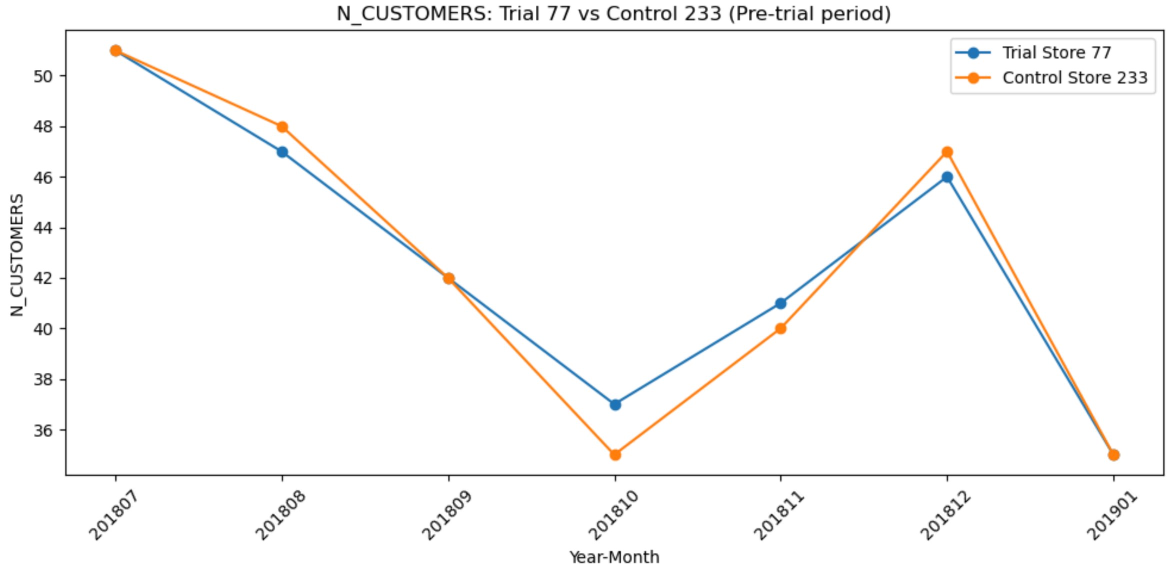

Weighted combination

In [31]:
```
def get_control_store_weighted(pretrial_df, trial_store):
    scores_sales = get_control_store(pretrial_df, "TOT_SALES", trial_store)[["ST
    scores_cust  = get_control_store(pretrial_df, "N_CUSTOMERS", trial_store)[["
    scores_txn   = get_control_store(pretrial_df, "TXN_PER_CUST", trial_store)[[

    combined = scores_sales.merge(scores_cust, on="STORE_NBR").merge(scores_txn,

    # zyada weight sales aur customers ko, kam weight noisy txn_per_cust ko
    combined["final_score"] = (
        0.5 * combined["score_sales"] +
        0.35 * combined["score_cust"] +
        0.15 * combined["score_txn"].fillna(0)
    )
    return combined.sort_values("final_score", ascending=False)

control_77_v2 = get_control_store_weighted(pretrial, 77)
print(control_77_v2.head(10))
```

```
C:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\_function_base_impl.py:2999:
RuntimeWarning: invalid value encountered in divide
  c /= stddev[:, None]
C:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\_function_base_impl.py:3000:
RuntimeWarning: invalid value encountered in divide
  c /= stddev[None, :]
    STORE_NBR  score_sales  score_cust  score_txn  final_score
0         233     0.951887    0.995179   0.221718     0.857514
3          17     0.864397    0.858222   0.684392     0.835235
4         115     0.815929    0.844780   0.623858     0.797217
9          84     0.758465    0.893020   0.670811     0.792411
1          41     0.880579    0.912850   0.186702     0.787793
5         167     0.814129    0.836580   0.493785     0.773935
20        145     0.689454    0.820516   0.782399     0.749267
10        254     0.754326    0.928783   0.313196     0.749217
7         234     0.796190    0.766486   0.539961     0.747360
11        111     0.748771    0.829074   0.459226     0.733445
```

In [32]:
```
plot_trial_vs_control(pretrial, "TOT_SALES", 77, 233)
plot_trial_vs_control(pretrial, "N_CUSTOMERS", 77, 233)
```




In [33]:
```
control_86_v2 = get_control_store_weighted(pretrial, 86)
print(control_86_v2.head(10))
```

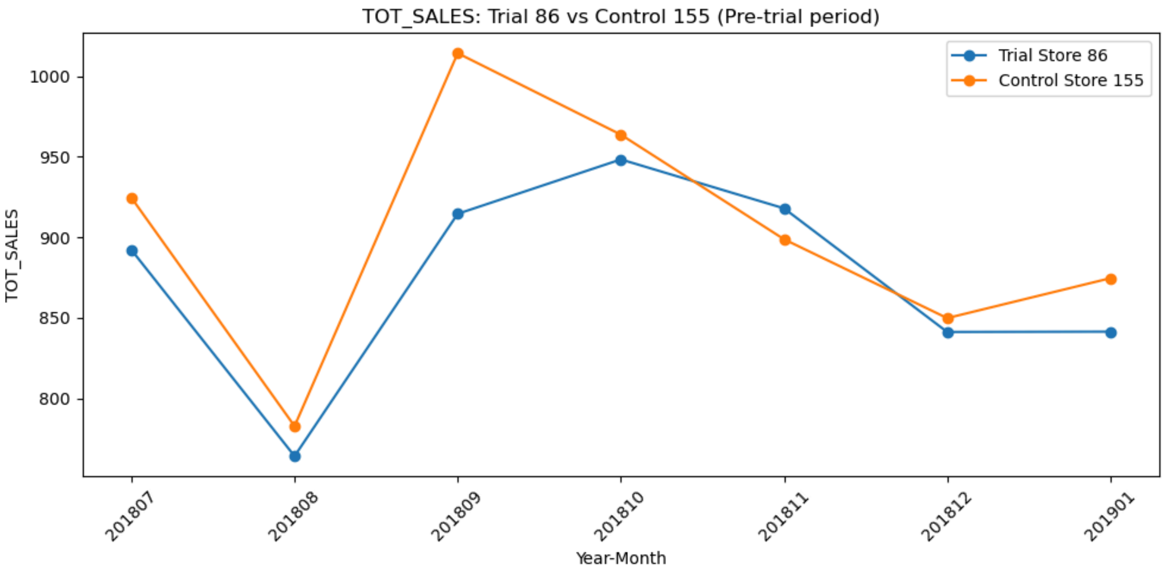
```
C:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\_function_base_impl.py:2999:
RuntimeWarning: invalid value encountered in divide
  c /= stddev[:, None]
C:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\_function_base_impl.py:3000:
RuntimeWarning: invalid value encountered in divide
  c /= stddev[None, :]
    STORE_NBR  score_sales  score_cust  score_txn  final_score
0         155     0.938423    0.971438   0.354823     0.862438
3         138     0.859483    0.845350   0.720934     0.833754
4         114     0.844751    0.901951   0.551526     0.820787
2         109     0.894150    0.875555   0.224814     0.787241
7         225     0.805183    0.857062   0.358176     0.756290
1         222     0.894964    0.695080   0.269302     0.731155
23         75     0.713052    0.755792   0.604752     0.711766
19        172     0.740062    0.759037   0.499904     0.710679
20        229     0.726716    0.727699   0.609789     0.709521
16         32     0.744380    0.636693   0.725273     0.703824
```

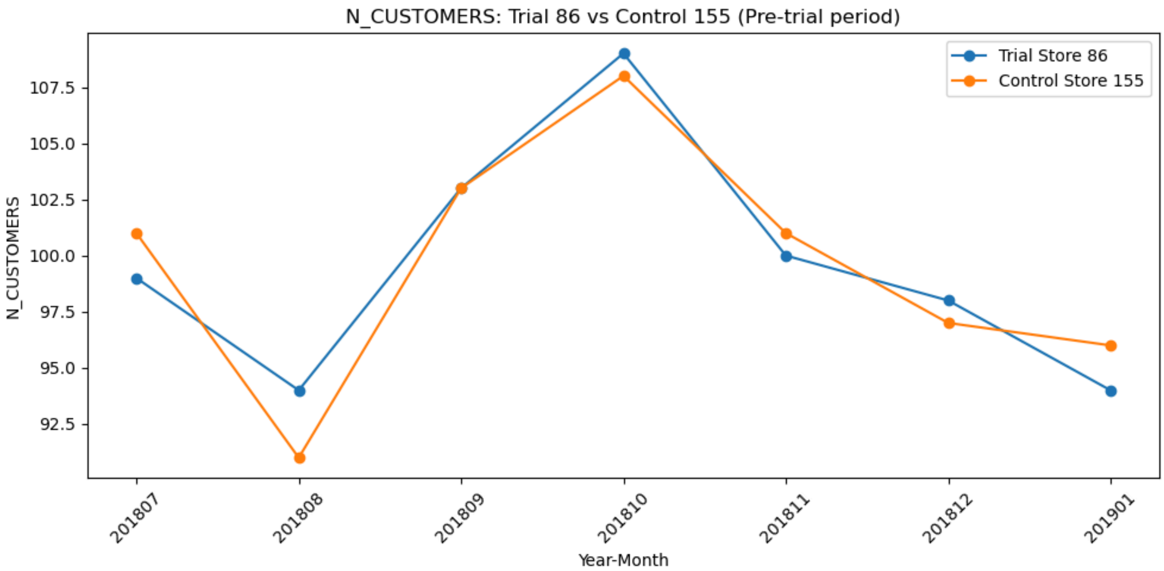
In [34]:
```
control_88_v2 = get_control_store_weighted(pretrial, 88)
print(control_88_v2.head(10))
```

```
C:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\_function_base_impl.py:2999:
RuntimeWarning: invalid value encountered in divide
  c /= stddev[:, None]
C:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\_function_base_impl.py:3000:
RuntimeWarning: invalid value encountered in divide
  c /= stddev[None, :]
```

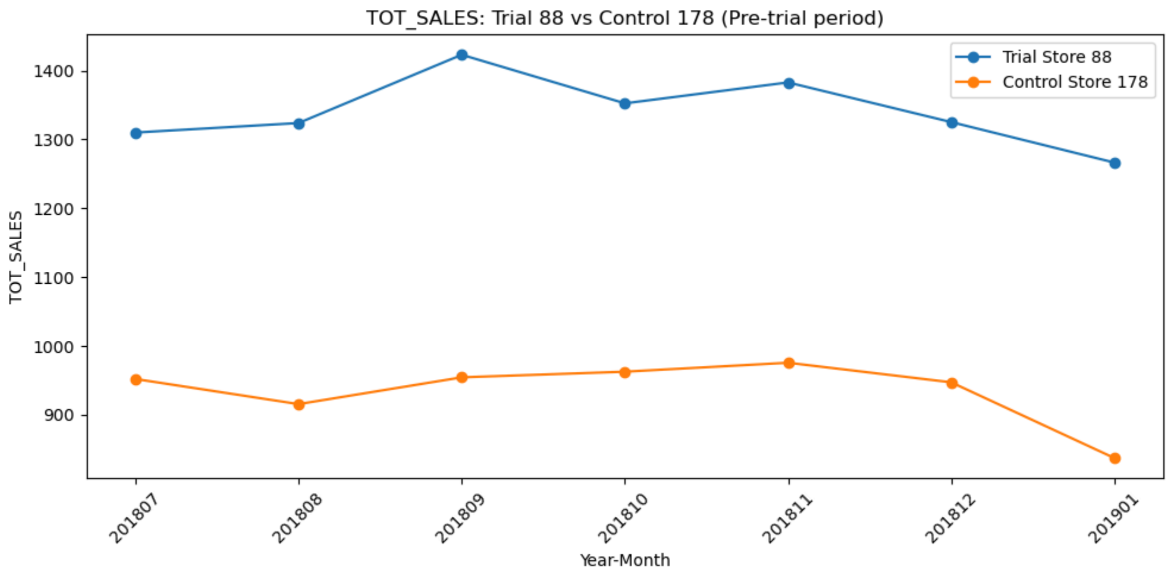
| | STORE_NBR | score_sales | score_cust | score_txn | final_score |
|---|---|---|---|---|---|
| 4 | 178 | 0.728884 | 0.884324 | 0.597168 | 0.763530 |
| 5 | 201 | 0.702425 | 0.721250 | 0.780996 | 0.720799 |
| 8 | 237 | 0.654240 | 0.973663 | 0.306244 | 0.713839 |
| 10 | 123 | 0.647417 | 0.766513 | 0.629791 | 0.686457 |
| 19 | 113 | 0.604013 | 0.826927 | 0.623202 | 0.684911 |
| 1 | 203 | 0.750972 | 0.619400 | 0.548776 | 0.674592 |
| 21 | 69 | 0.594328 | 0.847398 | 0.469808 | 0.664225 |
| 6 | 106 | 0.687683 | 0.527287 | 0.865123 | 0.658161 |
| 2 | 91 | 0.733564 | 0.436382 | 0.595418 | 0.608828 |
| 15 | 26 | 0.619799 | 0.630513 | 0.278717 | 0.572387 |

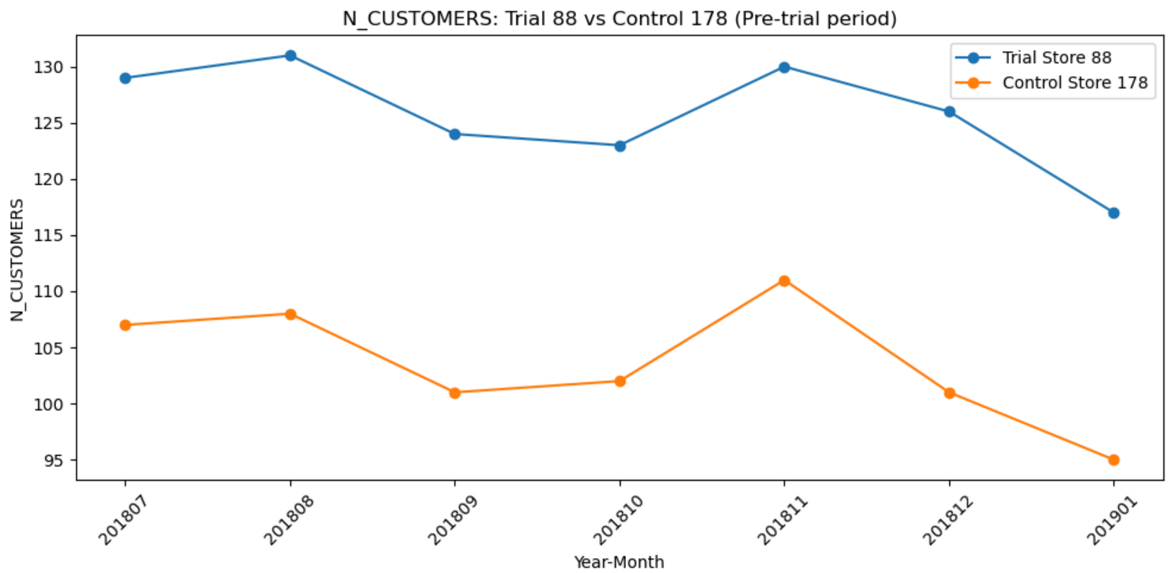
In [37]:
```
plot_trial_vs_control(pretrial, "TOT_SALES", 86, 155)
plot_trial_vs_control(pretrial, "N_CUSTOMERS", 86, 155)
```




In [38]:
```
plot_trial_vs_control(pretrial, "TOT_SALES", 88, 178)
plot_trial_vs_control(pretrial, "N_CUSTOMERS", 88, 178)
```

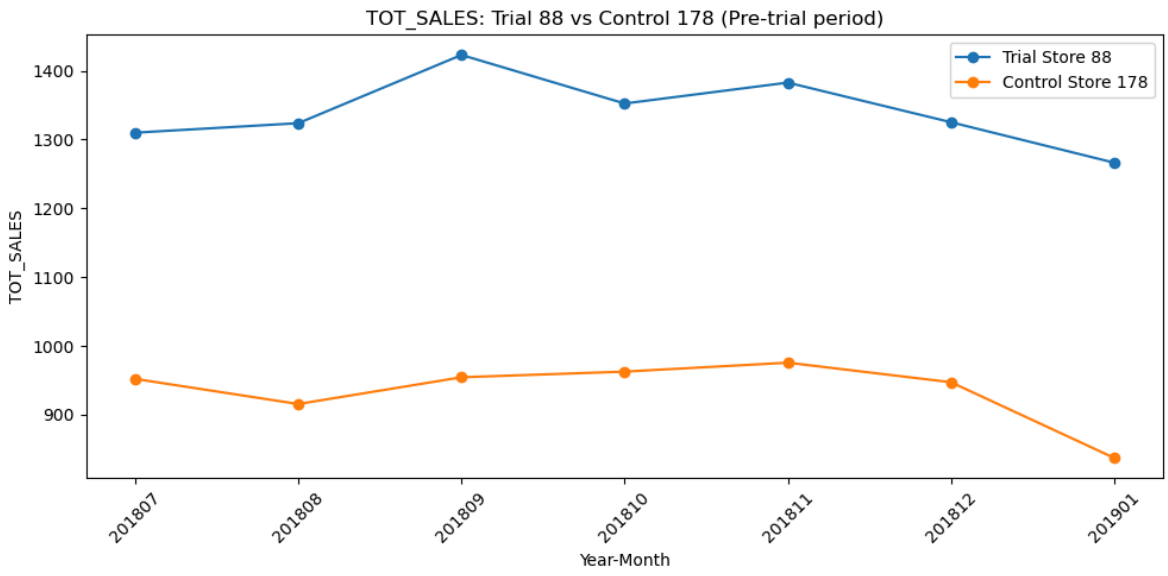


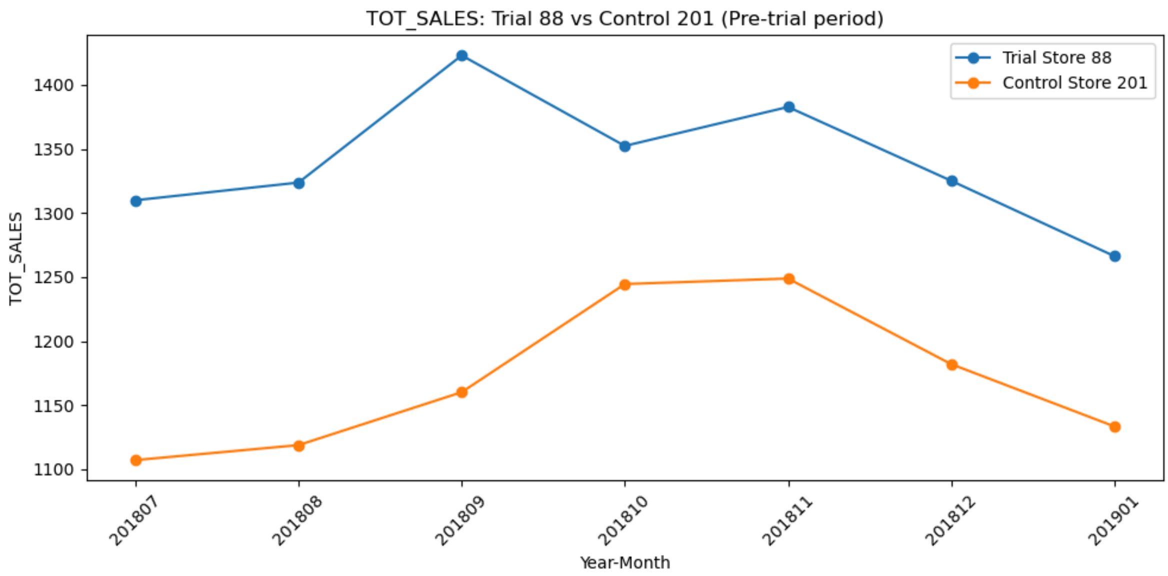



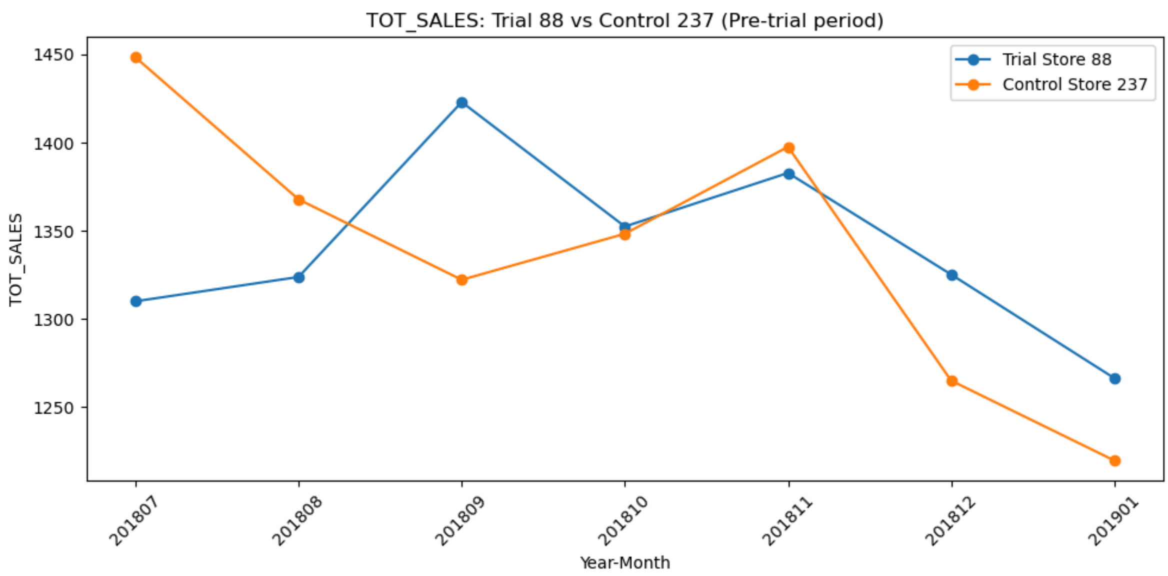

In [39]:
```
control_88_v2 = get_control_store_weighted(pretrial, 88)
print(control_88_v2.head(8))
```

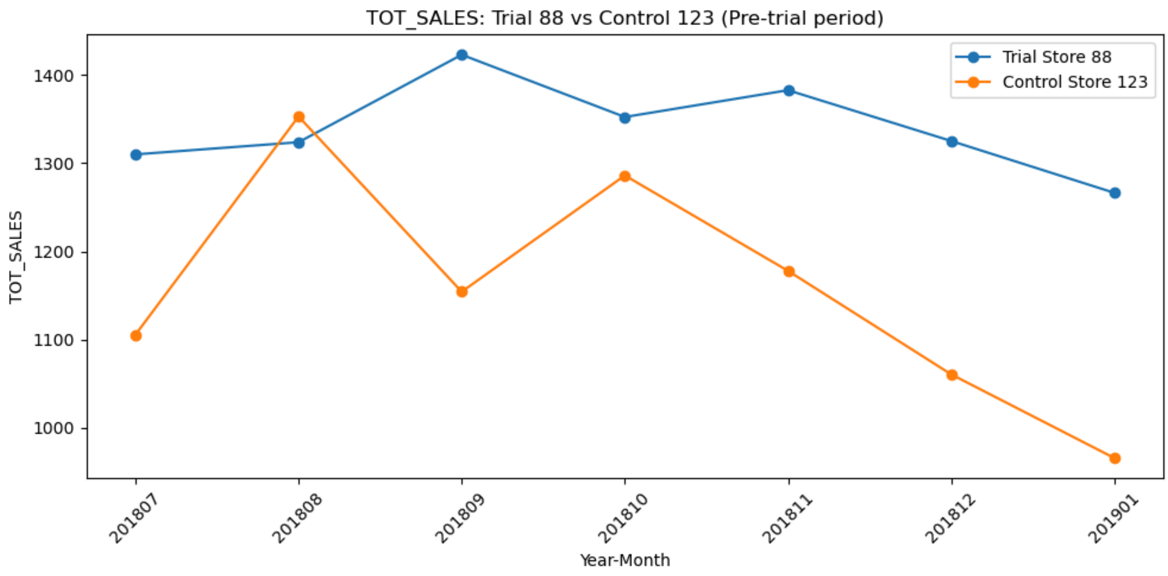
```
C:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\_function_base_impl.py:2999:
RuntimeWarning: invalid value encountered in divide
  c /= stddev[:, None]
C:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\_function_base_impl.py:3000:
RuntimeWarning: invalid value encountered in divide
  c /= stddev[None, :]
    STORE_NBR  score_sales  score_cust  score_txn  final_score
4         178     0.728884    0.884324   0.597168     0.763530
5         201     0.702425    0.721250   0.780996     0.720799
8         237     0.654240    0.973663   0.306244     0.713839
10        123     0.647417    0.766513   0.629791     0.686457
19        113     0.604013    0.826927   0.623202     0.684911
1         203     0.750972    0.619400   0.548776     0.674592
21         69     0.594328    0.847398   0.469808     0.664225
6         106     0.687683    0.527287   0.865123     0.658161
```

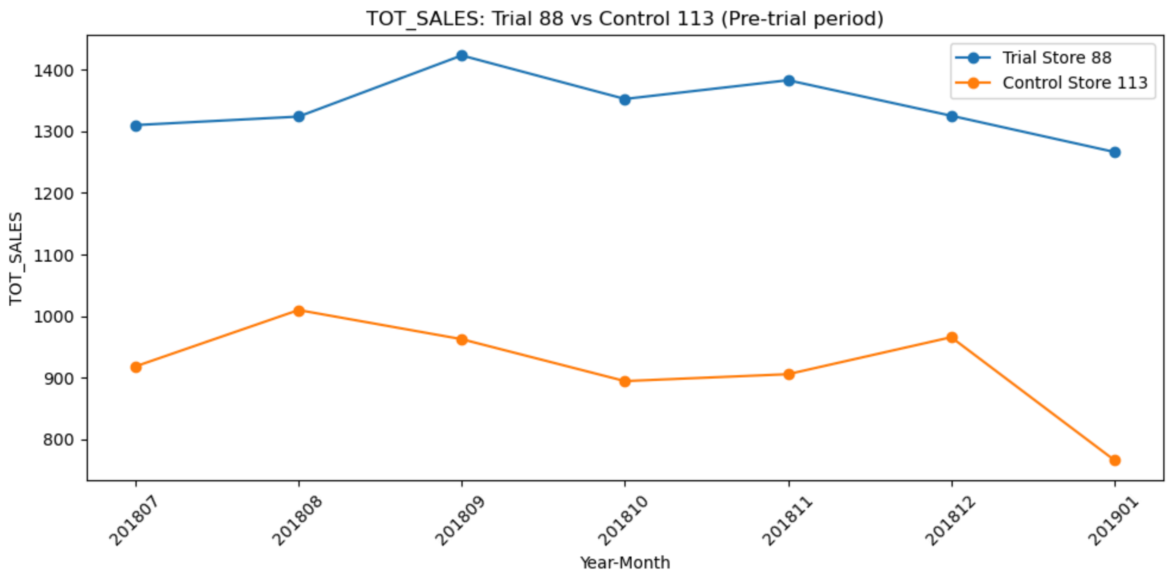
In [41]:
```
for candidate in control_88_v2["STORE_NBR"].head(5):
    plot_trial_vs_control(pretrial, "TOT_SALES", 88, candidate)
```


TOT_SALES: Trial 88 vs Control 178 (Pre-trial period)
Trial Store 88
Control Store 178
TOT_SALES
Year-Month
201807
201808
201809
201810
201811
201812
201901


TOT_SALES: Trial 88 vs Control 201 (Pre-trial period)
Trial Store 88
Control Store 201
TOT_SALES
Year-Month
201807
201808
201809
201810
201811
201812
201901


TOT_SALES: Trial 88 vs Control 237 (Pre-trial period)
Trial Store 88
Control Store 237
TOT_SALES
Year-Month
201807
201808
201809
201810
201811
201812
201901

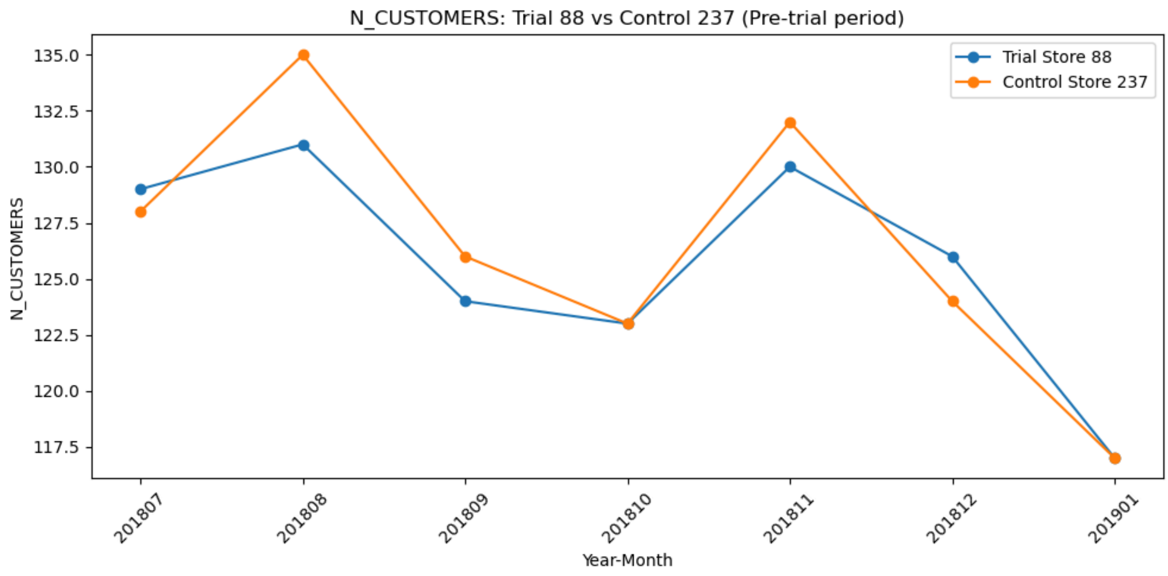



In [42]:
```
# Check karo store 88 average sales kitna hai vs candidates
for s in [88, 178, 201, 237, 123, 113]:
    avg = pretrial[pretrial["STORE_NBR"]==s]["TOT_SALES"].mean()
    print(f"Store {s}: avg pretrial sales = {avg:.1f}")
```

```
Store 88: avg pretrial sales = 1340.5
Store 178: avg pretrial sales = 934.9
Store 201: avg pretrial sales = 1170.8
Store 237: avg pretrial sales = 1338.4
Store 123: avg pretrial sales = 1157.5
Store 113: avg pretrial sales = 917.6
```

In [44]:
```
plot_trial_vs_control(pretrial, "N_CUSTOMERS", 88, 237)
```




In [45]:
```
trial_months = [201902, 201903, 201904]

trial_period = monthly_metrics_full[monthly_metrics_full["YEARMONTH"].isin(trial

def compare_trial_vs_control(trial_period_df, trial_store, control_store, metric
    trial_data = trial_period_df[trial_period_df["STORE_NBR"]==trial_store].sort
    control_data = trial_period_df[trial_period_df["STORE_NBR"]==control_store].

    comparison = pd.DataFrame({
        "YEARMONTH": trial_data["YEARMONTH"].values,
        f"Trial_{trial_store}": trial_data[metric_col].values,
        f"Control_{control_store}": control_data[metric_col].values,
    })
    comparison["pct_uplift"] = (
        (comparison[f"Trial_{trial_store}"] - comparison[f"Control_{control_stor
        / comparison[f"Control_{control_store}"] * 100
    )
    return comparison

result_77 = compare_trial_vs_control(trial_period, 77, 233, "TOT_SALES")
print(result_77)
```

```
   YEARMONTH  Trial_77  Control_233  pct_uplift
0     201902     235.0        244.0   -3.688525
1     201903     278.5        199.1   39.879458
2     201904     263.5        158.6   66.141236
```

In [46]:
```
def calc_scaling_factor(pretrial_df, trial_store, control_store, metric_col="TOT
    trial_avg = pretrial_df[pretrial_df["STORE_NBR"]==trial_store][metric_col].m
    control_avg = pretrial_df[pretrial_df["STORE_NBR"]==control_store][metric_co
    return trial_avg / control_avg

scale_77 = calc_scaling_factor(pretrial, 77, 233, "TOT_SALES")
print("Scaling factor:", scale_77)
```

```
Scaling factor: 1.0236173032895532
```

In [47]:
```
def get_pretrial_pct_diff(pretrial_df, trial_store, control_store, metric_col="T
    scale = calc_scaling_factor(pretrial_df, trial_store, control_store, metric_

    trial_data = pretrial_df[pretrial_df["STORE_NBR"]==trial_store].sort_values(
    control_data = pretrial_df[pretrial_df["STORE_NBR"]==control_store].sort_val
```

```python
    scaled_control = control_data[metric_col].values * scale
    pct_diff = (trial_data[metric_col].values - scaled_control) / scaled_control

    return pct_diff

pct_diff_77 = get_pretrial_pct_diff(pretrial, 77, 233, "TOT_SALES")
print("Pre-trial % differences (baseline noise):", pct_diff_77)
print("Std dev of baseline noise:", pct_diff_77.std())
```

```
Pre-trial % differences (baseline noise): [ -0.25727107 -12.69499722  -3.76023774
7.58303453  13.2515791
  -6.67164093  12.4980295 ]
Std dev of baseline noise: 9.219915451817057
```

In [48]:

```python
from scipy import stats

std_dev = pct_diff_77.std()
degrees_freedom = len(pct_diff_77) - 1  # 7 pretrial months - 1 = 6

result_77["t_value"] = result_77["pct_uplift"] / std_dev

# 95% one-tailed critical value (hum check kar rahe hain trial > control, isliye
t_critical = stats.t.ppf(0.95, degrees_freedom)
print("t-critical value (95% confidence):", t_critical)

print(result_77[["YEARMONTH", "pct_uplift", "t_value"]])
```

```
t-critical value (95% confidence): 1.9431802805153022
   YEARMONTH  pct_uplift   t_value
0     201902   -3.688525 -0.400061
1     201903   39.879458  4.325360
2     201904   66.141236  7.173736
```

In [50]:

```python
def full_trial_analysis(pretrial_df, trial_period_df, trial_store, control_store
    scale = calc_scaling_factor(pretrial_df, trial_store, control_store, metric_
    pct_diff = get_pretrial_pct_diff(pretrial_df, trial_store, control_store, me
    std_dev = pct_diff.std()
    dof = len(pct_diff) - 1
    t_critical = stats.t.ppf(0.95, dof)

    result = compare_trial_vs_control(trial_period_df, trial_store, control_stor
    result["t_value"] = result["pct_uplift"] / std_dev
    result["significant"] = result["t_value"] > t_critical

    print(f"\n=== Trial Store {trial_store} vs Control Store {control_store} ({m
    print(f"Baseline noise (std dev): {std_dev:.2f}%  |  t-critical: {t_critical
    print(result[["YEARMONTH","pct_uplift","t_value","significant"]])
    return result

result_86 = full_trial_analysis(pretrial, trial_period, 86, 155, "TOT_SALES")
result_88 = full_trial_analysis(pretrial, trial_period, 88, 237, "TOT_SALES")
```

```
=== Trial Store 86 vs Control Store 155 (TOT_SALES) ===
Baseline noise (std dev): 3.49%  |  t-critical: 1.943
   YEARMONTH  pct_uplift   t_value  significant
0     201902    2.468582  0.707536        False
1     201903   27.647936  7.924353         True
2     201904    0.426237  0.122167        False

=== Trial Store 88 vs Control Store 237 (TOT_SALES) ===
Baseline noise (std dev): 5.30%  |  t-critical: 1.943
   YEARMONTH  pct_uplift   t_value  significant
0     201902   -2.462984 -0.464689        False
1     201903   22.264526  4.200626         True
2     201904   19.491948  3.677526         True
```

In [51]:

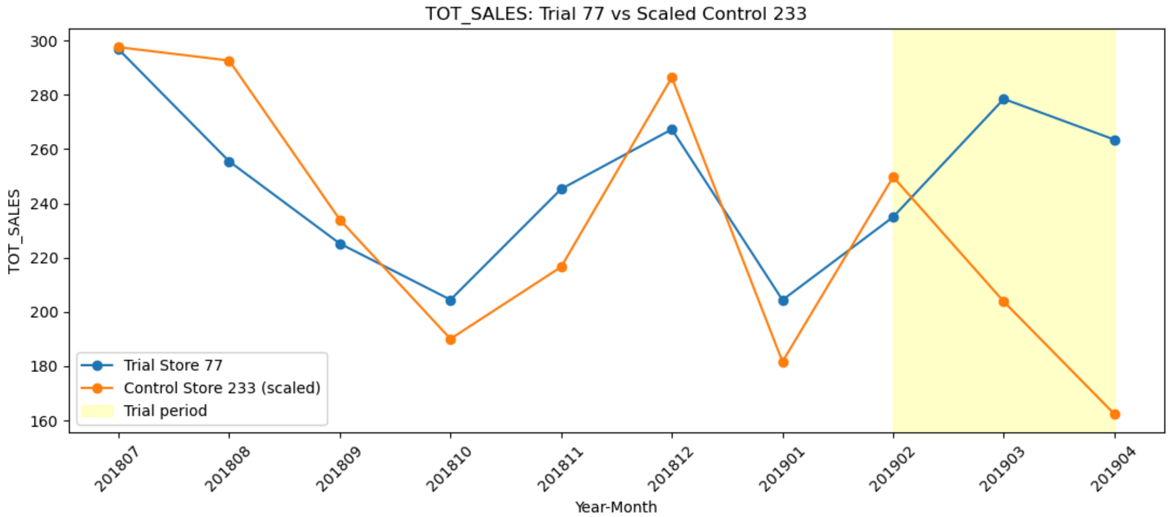
```python
result_77_cust = full_trial_analysis(pretrial, trial_period, 77, 233, "N_CUSTOME
result_86_cust = full_trial_analysis(pretrial, trial_period, 86, 155, "N_CUSTOME
result_88_cust = full_trial_analysis(pretrial, trial_period, 88, 237, "N_CUSTOME
```

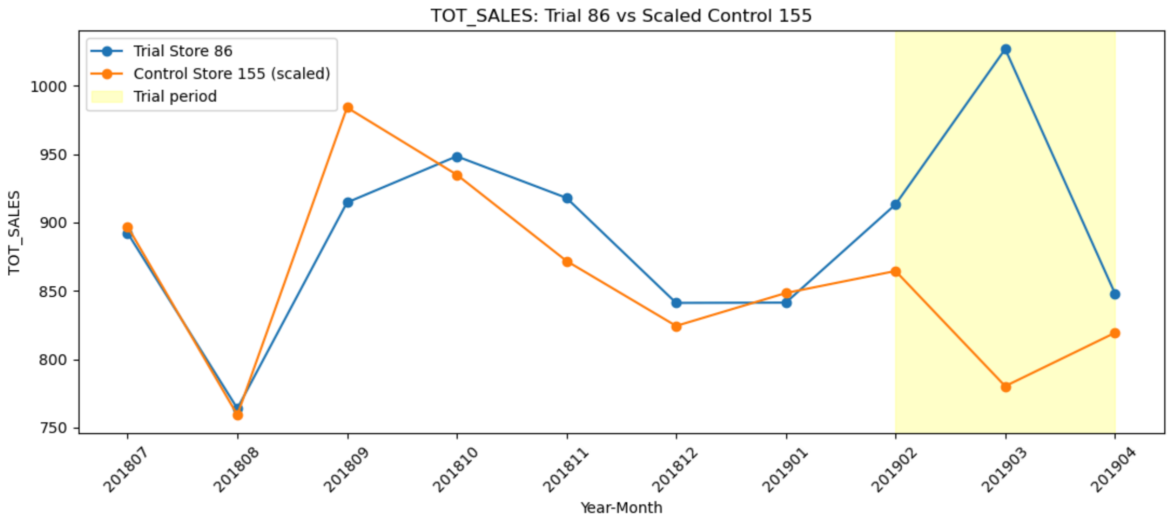
```
=== Trial Store 77 vs Control Store 233 (N_CUSTOMERS) ===
Baseline noise (std dev): 2.54%  |  t-critical: 1.943
   YEARMONTH  pct_uplift    t_value  significant
0     201902    0.000000   0.000000        False
1     201903   25.000000   9.840789         True
2     201904   56.666667  22.305789         True

=== Trial Store 86 vs Control Store 155 (N_CUSTOMERS) ===
Baseline noise (std dev): 1.77%  |  t-critical: 1.943
   YEARMONTH  pct_uplift    t_value  significant
0     201902   12.631579   7.121215         True
1     201903   22.340426  12.594702         True
2     201904    6.060606   3.416745         True

=== Trial Store 88 vs Control Store 237 (N_CUSTOMERS) ===
Baseline noise (std dev): 1.47%  |  t-critical: 1.943
   YEARMONTH  pct_uplift   t_value  significant
0     201902   -1.587302 -1.082520        False
1     201903   12.605042  8.596483         True
2     201904    6.666667  4.546584         True
```

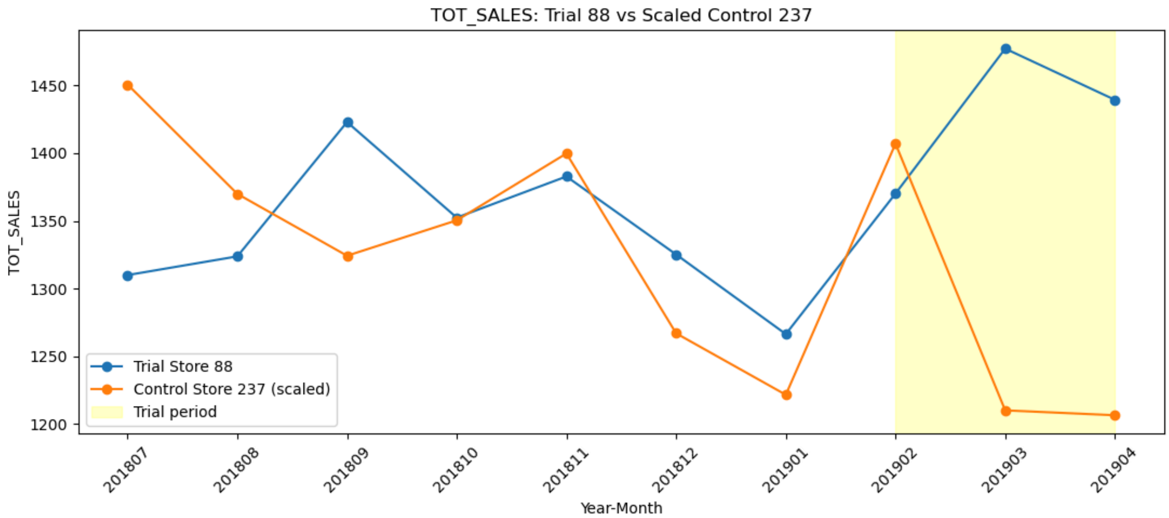
In [52]:

```python
def plot_full_trial_chart(pretrial_df, trial_df, trial_store, control_store, met
    scale = calc_scaling_factor(pretrial_df, trial_store, control_store, metric_

    full_data = pd.concat([pretrial_df, trial_df])
    trial_data = full_data[full_data["STORE_NBR"]==trial_store].sort_values("YEA
    control_data = full_data[full_data["STORE_NBR"]==control_store].sort_values(

    plt.figure(figsize=(11,5))
    plt.plot(trial_data["YEARMONTH"].astype(str), trial_data[metric_col], marker
    plt.plot(control_data["YEARMONTH"].astype(str), control_data[metric_col]*sca
    plt.axvspan(7, 9, color="yellow", alpha=0.2, label="Trial period")  # Feb-Ap
    plt.title(f"{metric_col}: Trial {trial_store} vs Scaled Control {control_sto
    plt.xlabel("Year-Month")
    plt.ylabel(metric_col)
    plt.legend()
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()

plot_full_trial_chart(pretrial, trial_period, 77, 233, "TOT_SALES")
```

```python
plot_full_trial_chart(pretrial, trial_period, 86, 155, "TOT_SALES")
plot_full_trial_chart(pretrial, trial_period, 88, 237, "TOT_SALES")
```




# Task 2 — Key Findings & Recommendation

**Methodology:**

- Control stores selected using a combined score (50% sales correlation/magnitude, 35% customer count, 15% avg transactions/customer) across the 7-month pre-trial period (Jul 2018 - Jan 2019), validated visually before finalising.

- Control store pairs: Trial 77 ↔ Control 233, Trial 86 ↔ Control 155, Trial 88 ↔ Control 237
- Trial period (Feb-Apr 2019) sales were compared to scaled control store sales using a t-test against pre-trial baseline variation (95% confidence).

**Results by store:**

*Store 77:* Significant sales uplift in March (+40%) and April (+66%), driven entirely by more customers visiting the store (customer count uplift matches the same months). No effect seen in February — likely an adjustment period.

*Store 86:* Significant sales uplift only in March (+28%). Customer numbers were up in all three trial months, but average spend per customer dropped in February and April, offsetting the extra footfall. This suggests the layout attracted visits but did not consistently convert to extra purchases in this store.

*Store 88:* Significant sales uplift in March (+22%) and April (+19%), consistent with increased customer numbers in the same months — a clean, customer-driven uplift similar to Store 77.

**Overall recommendation for Julia:** 2 of 3 trial stores (77 and 88) show a strong, consistent, statistically significant sales uplift driven by increased customer visits, with the effect strengthening over time. Store 86 shows a weaker and less consistent result, suggesting the layout's success may depend on store-specific factors (e.g. local customer base, store size, or execution consistency).

We recommend a broader rollout of the trial layout, with store 86's result flagged for further investigation (e.g. was the layout implemented correctly there, or does this store's customer base behave differently) before committing to a full chain-wide rollout.

In [ ]:
```

```